

# Series Solutions of Differential Equations

Prepared by Staples Education

*Unit 13: Power Series, the Frobenius Method, and the Classical Special Functions*

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### Unit 13 at a Glance: Series Solutions

Most differential equations have no closed-form elementary solution. The series method assumes the solution is an (possibly singular) analytic function  $y = \sum a_n x^n$ , substitutes it into the equation, and matches coefficients of each power of  $x$  to extract a recurrence for the  $a_n$ . At an ordinary point a plain power series works; at a regular singular point the Frobenius factor  $x^r$  is required. Applied to the great equations of mathematical physics, the method manufactures the classical special functions.

- **Taylor foundations** — the coefficient formula  $f^{(n)}(a)/n!$ , the standard library ( $e^x$ ,  $\sin$ ,  $\cos$ ,  $\frac{1}{1-x}$ ), and the radius of convergence.
- **Power series solutions** — ordinary points, the ansatz  $y = \sum a_n x^n$ , re-indexing the derivatives, and reading off a recurrence.
- **The Frobenius method** — regular singular points, the ansatz  $x^r \sum a_n x^n$ , the indicial equation, and its three root cases.
- **Legendre and Bessel** — the two flagship equations, polynomial versus transcendental solutions, and orthogonality.
- **Hermite, Laguerre, Chebyshev** — three more orthogonal families, their weights, recurrences, and physical homes.

## 13.1 Taylor Series and the Exponential Function

*Taylor and Maclaurin expansions, the coefficient formula  $f^{(n)}(a)/n!$ , the standard library of series, and the radius of convergence via the ratio test.*

### Taylor / Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, \quad \text{Maclaurin: } a = 0.$$

### The Standard Library

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad (|x| < 1).$$

### Radius of Convergence (Ratio Test)

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|, \quad \text{series converges for } |x - a| < R.$$

### Method — Build a Maclaurin Series

1. Compute the derivatives  $f^{(n)}(0)$  at the center.
2. Form the coefficients  $a_n = f^{(n)}(0)/n!$ .
3. Assemble the series  $\sum_{n \geq 0} a_n x^n$ .
4. Determine the radius of convergence with the ratio test.

### Worked Example — Maclaurin Series of $e^x$

Build the series for  $f(x) = e^x$  about  $x = 0$  and read off the cubic polynomial.

$$f^{(n)}(x) = e^x \Rightarrow f^{(n)}(0) = 1 \text{ for all } n$$

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{n!}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \frac{1/(n+1)!}{1/n!} = \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0 \Rightarrow R = \infty$$

### Key Takeaway

An **analytic function** equals its own **Taylor series** inside the radius of convergence  $R$ . This is the licence for the entire unit: if a solution is analytic, positing  $y = \sum a_n x^n$  loses nothing.

### Common Pitfall

A Taylor series only represents  $f$  *inside* its radius of convergence. The geometric series  $\sum x^n$  equals  $\frac{1}{1-x}$  only for  $|x| < 1$ ; plugging in  $x = 2$  is meaningless even though  $\frac{1}{1-x}$  is perfectly finite there.

## 13.2 Power Series Solutions at an Ordinary Point

Ordinary points, the ansatz  $y = \sum a_n x^n$ , term-by-term differentiation and re-indexing, and extracting a recurrence relation for the coefficients.

### The Ansatz and Its Derivatives (re-indexed to $x^n$ )

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=0}^{\infty} (n+1)a_{n+1}x^n, \quad y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n.$$

### Ordinary Point Test

For  $y'' + p(x)y' + q(x)y = 0$ ,  $x_0$  is ordinary  $\iff p, q$  are both analytic at  $x_0$ .

### Guaranteed Radius

$R \geq$  distance from  $x_0$  to the nearest singular point in the complex plane.

### Method — Series Solution at an Ordinary Point

1. Confirm  $x_0$  is an ordinary point ( $p, q$  analytic there).
2. Substitute  $y = \sum a_n x^n$  and the re-indexed derivatives into the equation.
3. Shift indices so every term carries the common power  $x^n$ .
4. Match coefficients of  $x^n$  to obtain a recurrence relation.
5. Solve the recurrence;  $a_0$  and  $a_1$  remain free and generate the two independent solutions.

### Worked Example — Solve $y'' + y = 0$ by Series

Find a power series solution about the ordinary point  $x = 0$  and identify the result.

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n$$

$$y'' + y = \sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n]x^n = 0$$

$$a_{n+2} = \frac{-a_n}{(n+2)(n+1)}$$

$$a_0 : 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \cos x; \quad a_1 : x - \frac{x^3}{3!} + \cdots = \sin x$$

$$y = a_0 \cos x + a_1 \sin x$$

### Key Takeaway

At an ordinary point the two free constants  $a_0$  and  $a_1$  are precisely the two arbitrary constants of a second-order ODE — the even-indexed and odd-indexed coefficients build the two independent solutions.

### Common Pitfall

You must *re-index* every series to the same power  $x^n$  before matching coefficients. Skipping the index shift after differentiating (forgetting the  $(n+1)$  or  $(n+2)(n+1)$  factor) is the single most common source of a wrong recurrence.

## 13.3 The Frobenius Method

*Regular singular points, the Frobenius ansatz  $x^r \sum a_n x^n$ , the indicial equation  $r(r-1) + p_0 r + q_0 = 0$ , and the three cases of its roots.*

### Regular Singular Point

$x_0$  is singular, yet  $(x - x_0)p(x)$  and  $(x - x_0)^2 q(x)$  are analytic at  $x_0$ .

### Frobenius Ansatz and Indicial Equation

$$y = x^r \sum_{n=0}^{\infty} a_n x^n, \quad r(r-1) + p_0 r + q_0 = 0, \quad p_0 = \lim_{x \rightarrow 0} x p(x), \quad q_0 = \lim_{x \rightarrow 0} x^2 q(x).$$

### The Three Root Cases ( $r_1 \geq r_2$ )

- (i)  $r_1 - r_2 \notin \mathbb{Z}$  : two clean Frobenius series.
- (ii)  $r_1 = r_2$  : second solution needs a  $\ln x$  term.
- (iii)  $r_1 - r_2 \in \mathbb{Z}^+$  :  $\ln x$  may appear.

### Method — The Frobenius Procedure

1. Confirm  $x_0$  is a regular singular point ( $xp$  and  $x^2q$  analytic).
2. Compute  $p_0 = \lim_{x \rightarrow 0} x p(x)$  and  $q_0 = \lim_{x \rightarrow 0} x^2 q(x)$ .
3. Form and solve the indicial equation  $r(r-1) + p_0 r + q_0 = 0$  for  $r_1 \geq r_2$ .
4. Substitute  $y = \sum a_n x^{n+r}$  with the larger root to get a recurrence for  $a_n$ .
5. Build the second solution according to the root case (a log term if roots coincide or differ by an integer).

### Worked Example — Indicial Roots of $x^2y'' + xy' - y = 0$

Classify  $x = 0$ , find  $p_0, q_0$ , and solve the indicial equation.

Standard form:  $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 0$ ,  $p(x) = \frac{1}{x}$ ,  $q(x) = -\frac{1}{x^2}$

$$p_0 = \lim_{x \rightarrow 0} x \cdot \frac{1}{x} = 1, \quad q_0 = \lim_{x \rightarrow 0} x^2 \cdot \left(-\frac{1}{x^2}\right) = -1$$

$$r(r-1) + (1)r + (-1) = r^2 - 1 = 0$$

$$r = 1 \text{ and } r = -1 \Rightarrow y = c_1x + c_2x^{-1}$$

### Key Takeaway

The Frobenius exponent  $x^r$  is exactly what captures the singular behavior a plain power series cannot. The indicial equation comes from the *lowest* power of  $x$  in the substitution, and its roots set the leading exponents of the two solutions.

### Common Pitfall

When the indicial roots are equal, or differ by a positive integer, the two series can collide. Do not assume two clean series — the smaller-root solution may require a **logarithmic term**  $y_1 \ln x$  for independence.

## 13.4 Legendre and Bessel Equations

*The two flagship equations of mathematical physics, the contrast between terminating polynomial solutions and transcendental ones, and orthogonality.*

### Legendre's Equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0, \quad \text{singular at } x = \pm 1, \text{ ordinary at } x = 0.$$

### Legendre Polynomials and Orthogonality

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1); \quad \int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}.$$

### Bessel's Equation of Order $\nu$

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0, \quad \text{regular singular at } x = 0, \text{ indicial roots } r = \pm \nu.$$

### Method — Recognize and Solve a Named Equation

1. Match the equation to its standard form and read off the parameter ( $n$  or  $\nu$ ).
2. Classify  $x = 0$ : Legendre is ordinary there (use a power series); Bessel is regular singular (use Frobenius).
3. For Legendre with integer  $n$ , the series terminates into the degree- $n$  polynomial  $P_n$ .
4. For Bessel, the larger root  $+\nu$  gives the bounded first-kind solution  $J_\nu$ .

### Worked Example — The Legendre Polynomial $P_2$

For  $n = 2$ , Legendre's equation  $(1 - x^2)y'' - 2xy' + 6y = 0$  has a terminating solution. Normalize so  $P_2(1) = 1$ .

$$\text{Even series: } a_2 = -\frac{n(n+1)}{2}a_0 = -\frac{6}{2}a_0 = -3a_0, \quad a_4 = 0 \text{ (terminates)}$$

$$y = a_0(1 - 3x^2)$$

$$\text{Normalize } P_2(1) = 1 : a_0(1 - 3) = -2a_0 = 1 \Rightarrow a_0 = -\frac{1}{2}$$

$$P_2(x) = -\frac{1}{2}(1 - 3x^2) = \frac{1}{2}(3x^2 - 1)$$

### Key Takeaway

The **special functions** are simply the named solutions of the physically important singular ODEs. Legendre arises from spherical symmetry and terminates into polynomials for integer  $n$ ; Bessel arises from cylindrical symmetry and gives the oscillatory  $J_\nu$ .

### Common Pitfall

Bessel's equation at  $x = 0$  is a *regular singular point*, not an ordinary point — the coefficients blow up there. A plain power series fails; you must use Frobenius with indicial roots  $r = \pm\nu$ .

## 13.5 Hermite, Laguerre, and Chebyshev Polynomials

*Three more classical orthogonal families: their differential equations, recurrences, weight functions, and the physical or numerical settings where each appears.*

### Hermite (physicists' convention)

$$y'' - 2xy' + 2ny = 0; \quad H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2; \quad \text{weight } e^{-x^2} \text{ on } (-\infty, \infty).$$

### Laguerre

$$xy'' + (1 - x)y' + ny = 0; \quad \text{weight } e^{-x} \text{ on } [0, \infty).$$

### Chebyshev (first kind)

$$(1-x^2)y'' - xy' + n^2y = 0; \quad T_n(\cos \theta) = \cos(n\theta)$$
$$T_0 = 1, \quad T_1 = x, \quad T_2 = 2x^2 - 1; \quad \text{weight } \frac{1}{\sqrt{1-x^2}}.$$

### Method — Place a Classical Family

1. Inspect the leading coefficient and the first-derivative term to fingerprint the equation.
2. Match to the family: constant lead with  $-2xy'$  is Hermite;  $xy''$  with  $(1-x)y'$  is Laguerre;  $(1-x^2)y''$  with a single  $xy'$  is Chebyshev.
3. Generate the polynomials from the family's recurrence.
4. Pair with the correct weight function for orthogonality.

### Worked Example — $H_2$ and $T_2$ from Recurrences

Use the Hermite recurrence  $H_{n+1} = 2xH_n - 2nH_{n-1}$  and the Chebyshev recurrence  $T_{n+1} = 2xT_n - T_{n-1}$  to build the degree-2 members.

$$H_2 = 2xH_1 - 2(1)H_0 = 2x(2x) - 2(1) = 4x^2 - 2$$

$$T_2 = 2xT_1 - T_0 = 2x(x) - 1 = 2x^2 - 1$$

$$\text{Contrast: } H_2 = 4x^2 - 2, \quad T_2 = 2x^2 - 1, \quad P_2 = \frac{1}{2}(3x^2 - 1)$$

### Key Takeaway

Each classical family is an **orthogonal basis under its own weight**, tailored to a geometry: Hermite ( $e^{-x^2}$ ) to the quantum harmonic oscillator, Laguerre ( $e^{-x}$ ) to the hydrogen atom, Chebyshev ( $1/\sqrt{1-x^2}$ ) to minimax approximation.

### Common Pitfall

Do not confuse the degree-2 members:  $H_2 = 4x^2 - 2$  (Hermite),  $T_2 = 2x^2 - 1$  (Chebyshev), and  $P_2 = \frac{1}{2}(3x^2 - 1)$  (Legendre) all look similar but come from different equations, recurrences, and weights.